

PEAK250 recurring precharge-module fault

A station-level hardware fault — HCU Precharge Module Fault_0x27 (vendor code 012900) — aborted **three consecutive customer charging sessions** within 41 minutes on 2026-07-26, each while power was already flowing. Service was restored by a remote hard reset, but the fault is a **degrading hardware condition** that recurs on charging and requires vendor inspection.

unit **A2111LHE0212606001**

firmware **ZD-SW4.2.4.1**

vendor **ZD Energy**

severity **Critical**

2026-07-26

01 Summary

Times are shown in UTC with site-local CEST (UTC+2) in parentheses; the charger reports in UTC.

- The dual-gun battery-buffered PEAK250 raised the identical fault **three times** — 09:14:47, 09:27:27 and 09:55:08 UTC (11:14, 11:27, 11:55 CEST).
- Each occurrence was a **station-level** fault (OCPP connector 0) that forced **both** guns to **Faulted** and aborted the active Gun 1 (CCS) session.
- All three happened **during charging** at 52–75 kW, not at plug-in, and under normal operating conditions (no overload).
- The time-to-fault shortened on each attempt — **2 min 40 s → 40 s → 30 s** — the signature of a degrading component.
- A separate, isolated **SECC sequence error** (vendor 163200) occurred earlier at 06:06 UTC on Gun 2 and did not recur.

Incident timeline

Reconstructed from the central-system durable fault log, the TimescaleDB meter store, and Loki. Each fault immediately followed an aborted charging session on Gun 1.

TIME (UTC / CEST)	SESSION	CHARGING BEFORE FAULT	FAULT	OUTCOME
09:14:47 11:14:47	tx 1021, Gun 1	2 min 40 s · peak 75 kW · EV SoC 38→42%	HCU Prechrg Module Fault_0x27 vendor 012900	Both guns → Faulted; session stopped (reason "Other"); ~11 min recovery
09:27:27 11:27:27	tx 1022, Gun 1	~40 s · peak 80 kW · same EV (SoC 42%)	HCU Prechrg Module Fault_0x27 vendor 012900	Both guns → Faulted; session stopped (reason "Other")
09:55:08 11:55:08	tx 1023, Gun 1	~30 s · peak 52 kW	HCU Prechrg Module Fault_0x27 vendor 012900	Both guns → Faulted; new customer session aborted
06:06:45 08:06:45	— (Gun 2 idle)	n/a	SECC detect the sequence error vendor 163200	Isolated; did not recur; Gun 1 charged normally 3 min later

For contrast, earlier sessions the same morning completed normally: tx 1019 (Gun 1, 08:09 CEST, 147 kW peak, ~35 min) and tx 1020 (Gun 2, 10:53 CEST). The fault is intermittent, not a hard failure of the whole unit.

Root-cause analysis

Four independent observations point to a defective, degrading component on the shared DC power stage.

SHARED

It downs both guns → shared component

The fault is reported on OCPP connector 0 (the station itself) and forces *both* charging points to **Faulted**. In the battery-buffered architecture both guns draw from one shared DC/power stage — consistent with the precharge module sitting on that shared bus rather than on a single gun.

MID-SESSION

It fires during charging, not at plug-in

Precharge normally runs once at session start to match the DC link to the EV battery before closing the main contactor. Here it faulted 2.5 minutes into a session with 62 kW already flowing — an unusual timing that suggests either continuous precharge-circuit monitoring detecting a fault, or a mid-session precharge re-run (e.g. on buffer↔grid source switching).

NORMAL LOAD

It occurs under normal conditions

At fault time the car drew 62–75 kW, buffer SoC was 94–100%, and grid infeed was ~40 kW — well within rating. Nothing was overloaded, which argues for a hardware defect rather than an operating-condition trip.

DEGRADING

Onset accelerates on each attempt

Time-to-fault fell from 2 min 40 s to 40 s to 30 s across the three attempts. A component that fails progressively sooner is the classic signature of thermal or wear-driven degradation, not a one-off transient.

Working hypothesis: a failing precharge module (contactor / resistor / DC-link monitoring) on the shared DC stage, degrading under repeated load. The precise meaning of internal sub-code **0x27** and vendor code **012900** requires ZD's confirmation.

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Customer impact

Three charging attempts were aborted mid-session. Signed OCMF meter data exists for each, so billing is defensible.

- **tx 1021** — 2.37 kWh delivered before abort; receipt #00000368.
- **tx 1022** — 0.20 kWh delivered (same driver retried); receipt #00000370.
- **tx 1023** — a new customer, ~30 s of charging before abort.

Each session ended with OCPP StopTransaction reason **Other** (fault-terminated), not **Local** (customer-ended). Affected drivers should be refunded to the actual energy delivered, and the pre-authorization holds released.

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Response & remediation

DONE

Remote hard reset — service restored

A hard reset was issued from our backend at 09:59:43 UTC and accepted by the charger. It power-cycled and returned with both guns **Available** and no fault state at 10:04 UTC — ~5 minutes total. A soft reset was deliberately avoided, as it would restart only the application and not clear a latched hardware fault.

DONE

Remote take-out-of-service capability added

The incident exposed that we had no way to remotely fence off faulty hardware. We wired the OCPP **ChangeAvailability** operation into the backend command surface (**POST** `/chargers/{id}/commands/change-availability`, Operative / Inoperative), so a faulting unit can now be marked Inoperative remotely pending inspection.

ACTION

Recommended: take the unit out of service

The reset clears the fault state but not the cause; the fault recurs on charging. Until ZD inspects, the unit (or at minimum Gun 1) should be set Inoperative so arriving customers see "unavailable" rather than a failed charge and a pre-auth hold.

For the vendor (ZD Energy)

Fault codes observed on unit A2111LHE0212606001, firmware ZD-SW4.2.4.1.

VENDOR CODE	TEXT	OCPP ERRORCODE	OCCURRENCES
012900	HCU Prechrg Module Fault_0x27	OtherError	3 × (09:14, 09:27, 09:55 UTC), all mid-session on Gun 1
163200	SECC detect the sequence error	OtherError	1 × (06:06 UTC), Gun 2, isolated

Questions for ZD:

- What does **012900 / HCU Prechrg Module Fault_0x27** indicate specifically — which component (precharge contactor, resistor, DC-link monitoring)?
- Why would a *precharge* fault trigger minutes into an active session, long after precharge completes? Is precharge re-run mid-session (e.g. on buffer↔grid source switching) in the buffered design?
- Why does the fault take **both** charging points down — is the precharge module shared across the two DC outputs?
- Does the accelerating onset (2 m 40 s → 40 s → 30 s) indicate a degrading / thermal condition versus a hard failure? Is the unit safe to keep in service?
- Recommended action — inspection or part replacement?
- Separately: is **163200 (SECC sequence error)** on Gun 2 related, or an independent vehicle-handshake issue?